Fratricide prevention depends on how systems (and people) work together

A functional hazard model of fratricide in an IAMD SoS

Matthew J. Squair, Amentum
Sam Mills, Australian Department of Defence

Australian System Safety Conference 2026, Adelaide

The ADF IAMD an enterprise system of systems



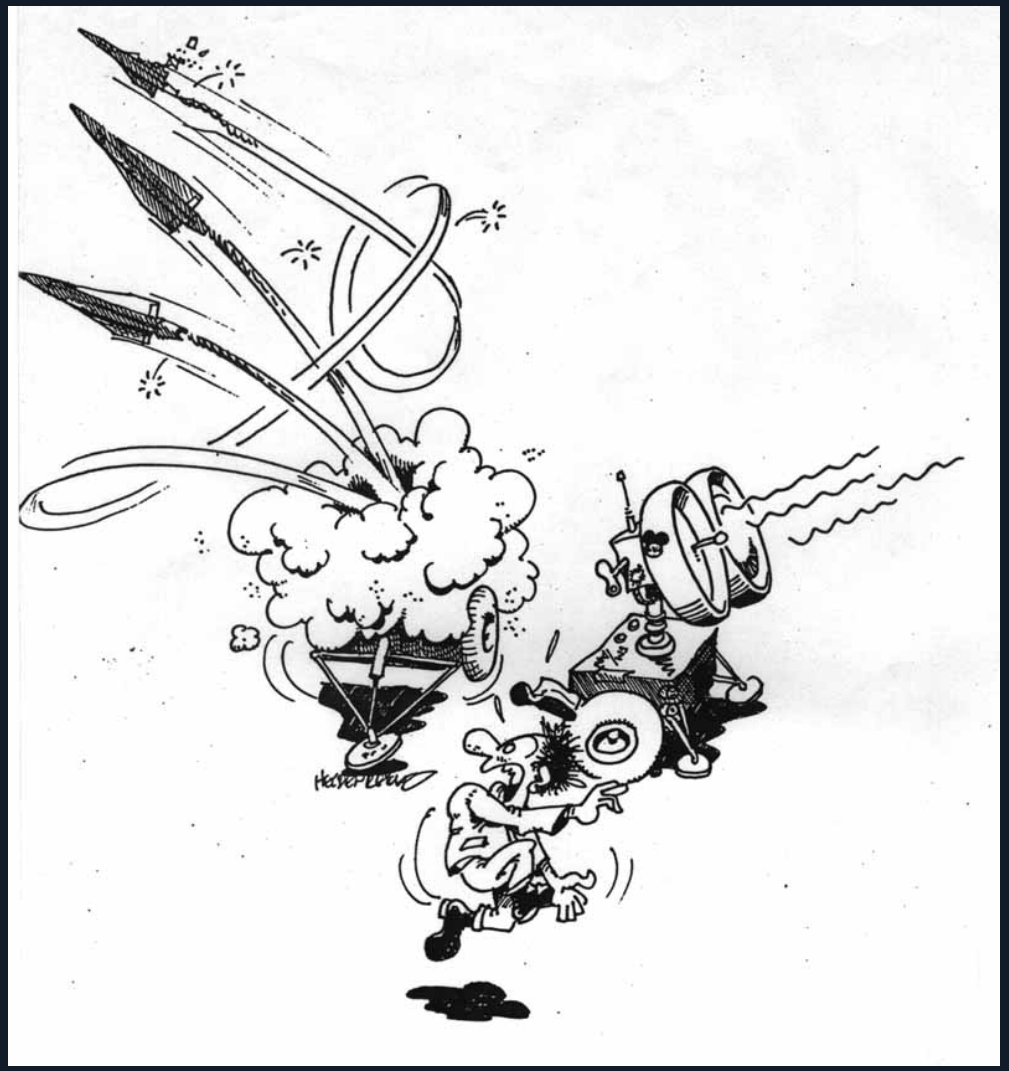


Joint Air Battle Management System





The integration problem



“But all the constituent systems passed their tests!”

How do we manage Fratricide risk in a SoS?

1. How do we model fratricide risk in a system of systems?

   (the analysis problem)

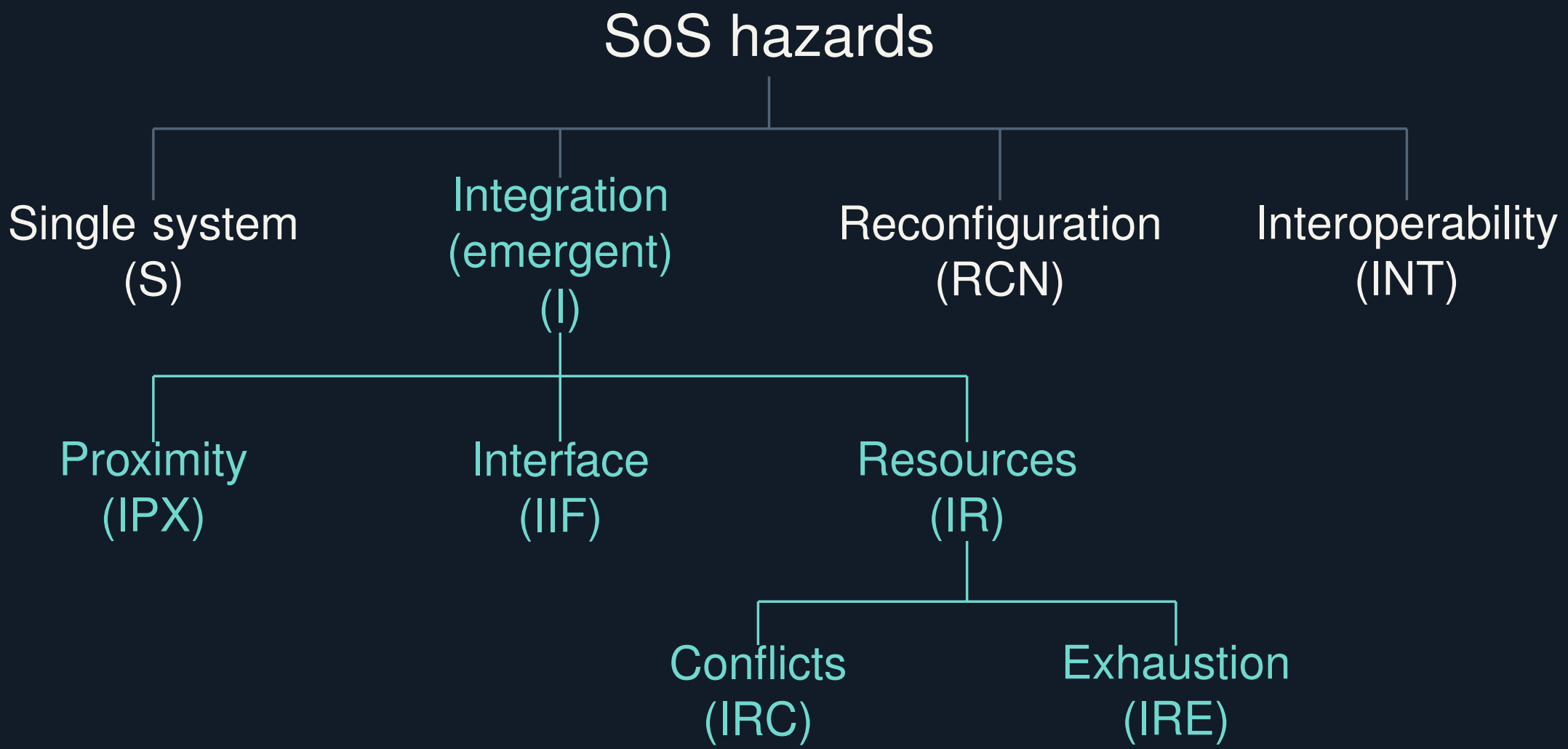
2. How do we then derive hazard controls from this model?

   (why we're doing this)

3. How do we know we're not drinking our own bath water?

   (completeness, correctness, etc.)

Let’s start with a taxonomy (Ericson’s)



Five interacting functional threads

Initial model

Tracking	Where is the contact?
Friend force tracking	Where are our aircraft?
Identification and classification	What is it?
Engagement	May we act, and can we stop?

Added from case studies

Airspace control	Can these activities coexist?

The SoS fratricide hazard (Swallom's SMO model)

a. Hazard source. A friendly platform occupies, or may enter, a region in which a friendly weapon engagement can expose it to harm. This condition can arise through intentional or unintentional sharing of airspace, irrespective of whether the platform is positively identified as friendly.

---

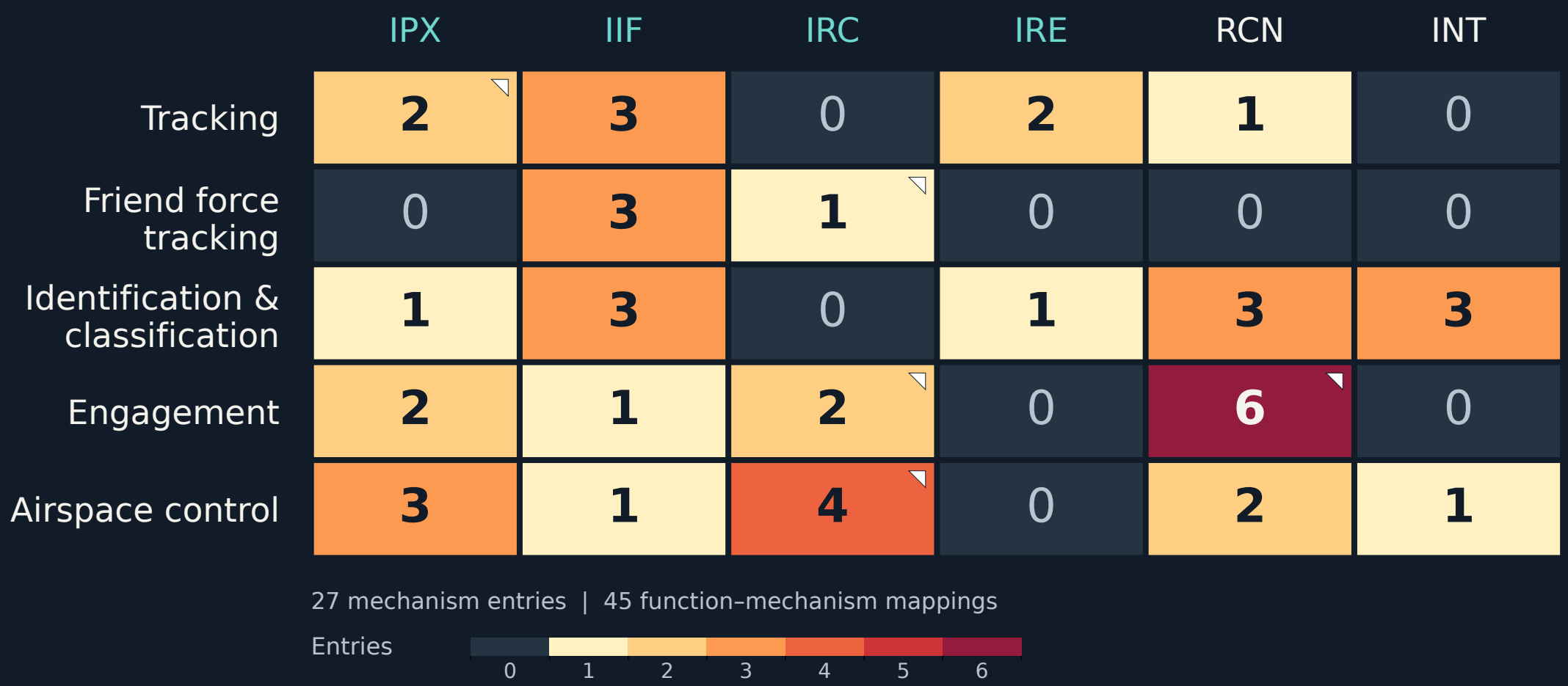
b. Causal mechanism. The SoS fails to maintain or apply the identity, position, airspace and engagement constraints needed to protect the platform. The platform may be selected as a target or acquired during an engagement directed at another contact.

---

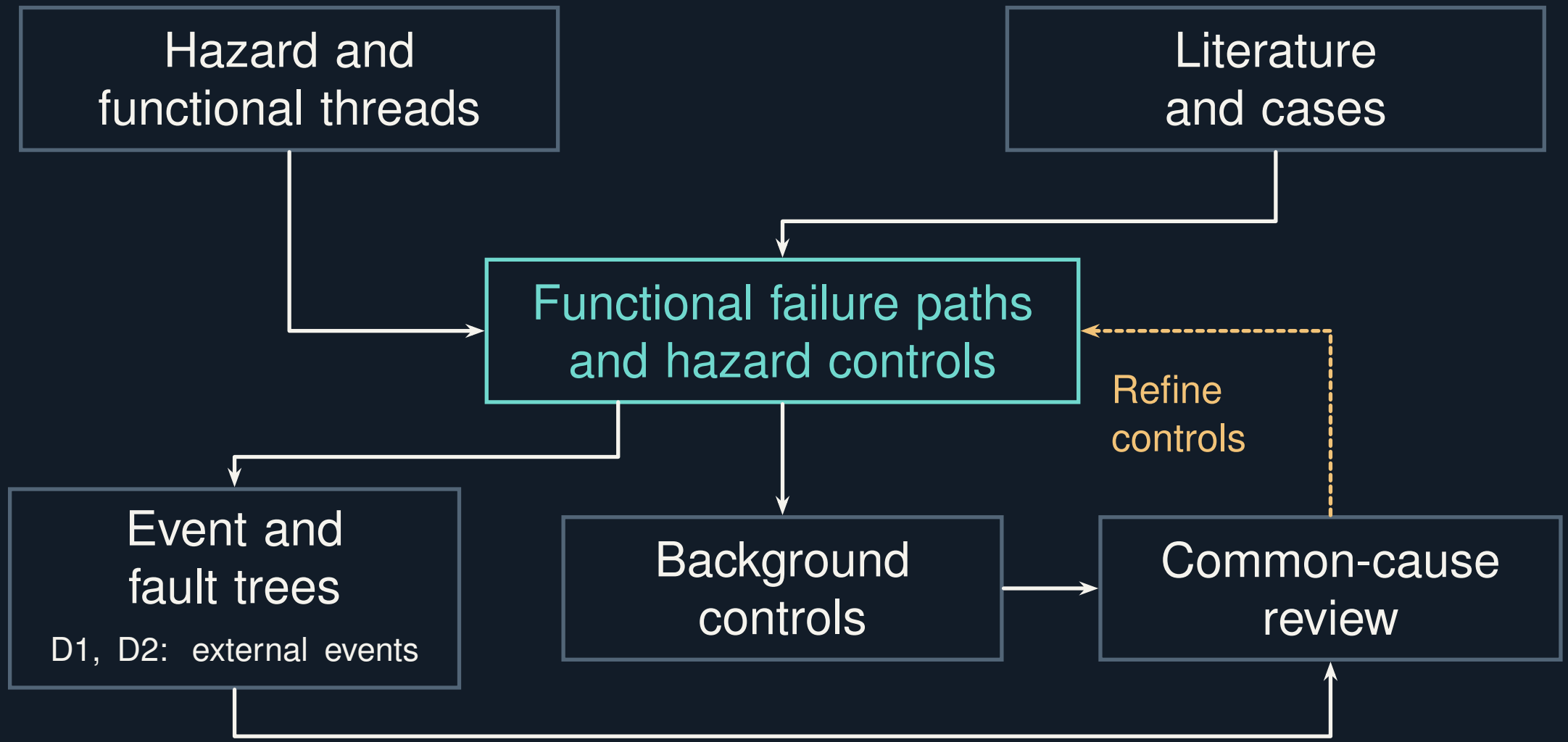
c. Outcome. The friendly platform is unintentionally hit or otherwise harmed by a friendly weapon.

Case studies modified the initial formulation.

Case studies were used to develop and test the functional model



Analysis Method – at least how we'd like to remember it



Opportunities for protection in the event sequence

Protection continues into the engagement

A	B	C1	C2	D1	D2
Airspace protection	Target-set exclusion	Inhibit launch	Intervene in time	Aircrew evasion	Weapon misses

Engineered protective barriers

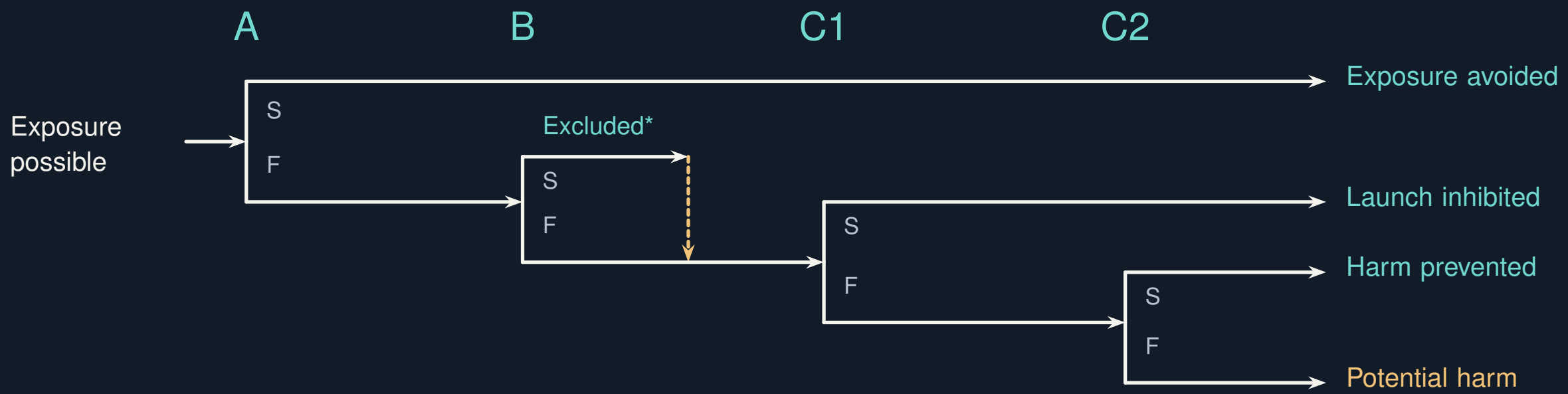
External events

Days / hours

Minutes / seconds

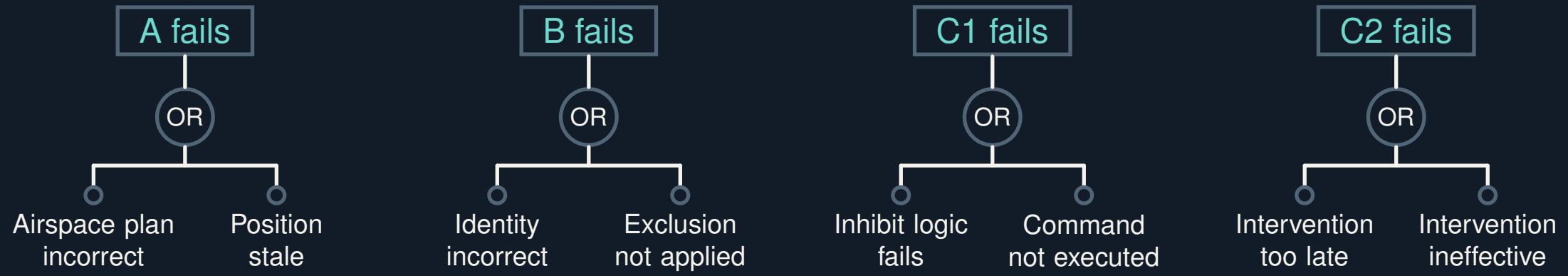
D1 and D2 can avert harm, but are not engineered protective barriers.

Illustrative event tree and supporting fault trees or obligatory slide that needs explaining



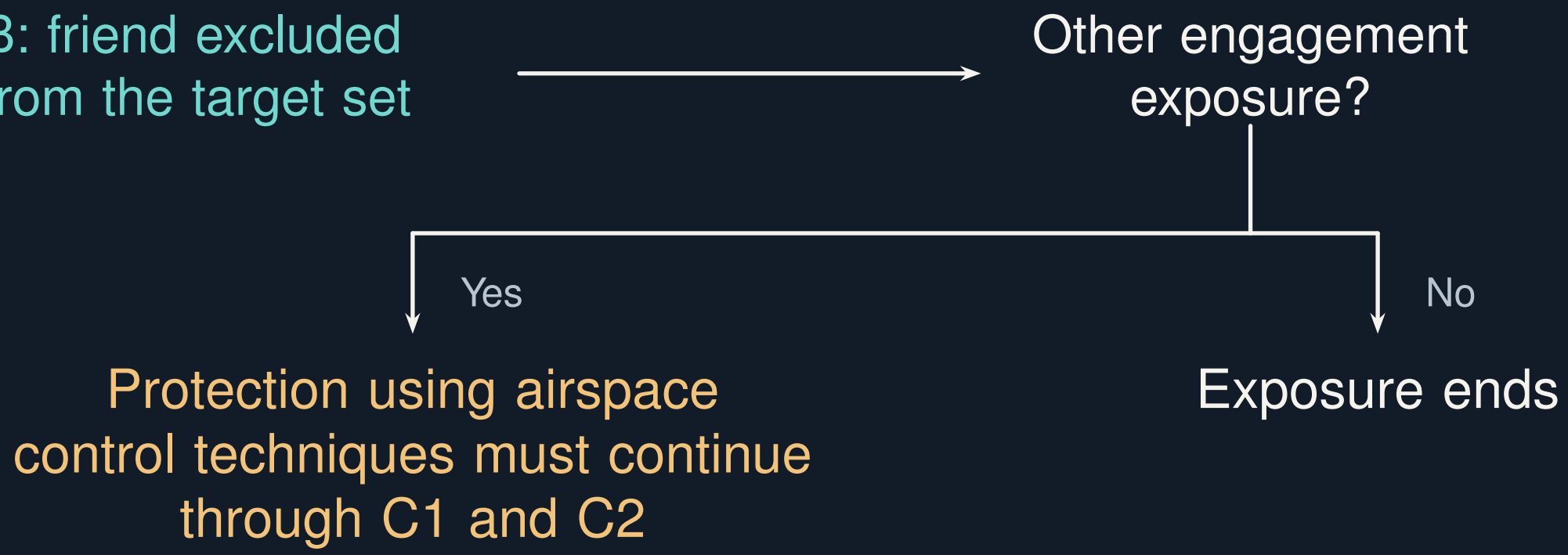
* If exposed to another engagement, continue through C1 and C2.

S: succeeds F: fails



Illustrative logic only: no probabilities or independence assumed; shared services can affect several barriers.

A friendly aircraft can remain exposed to an engagement aimed at another contact



Resulted in Dynamic separation: HC-44

Protect the aircraft from the seeker’s acquisition volume and the fly-out path

Before launch: C1

Withhold or revoke launch permission

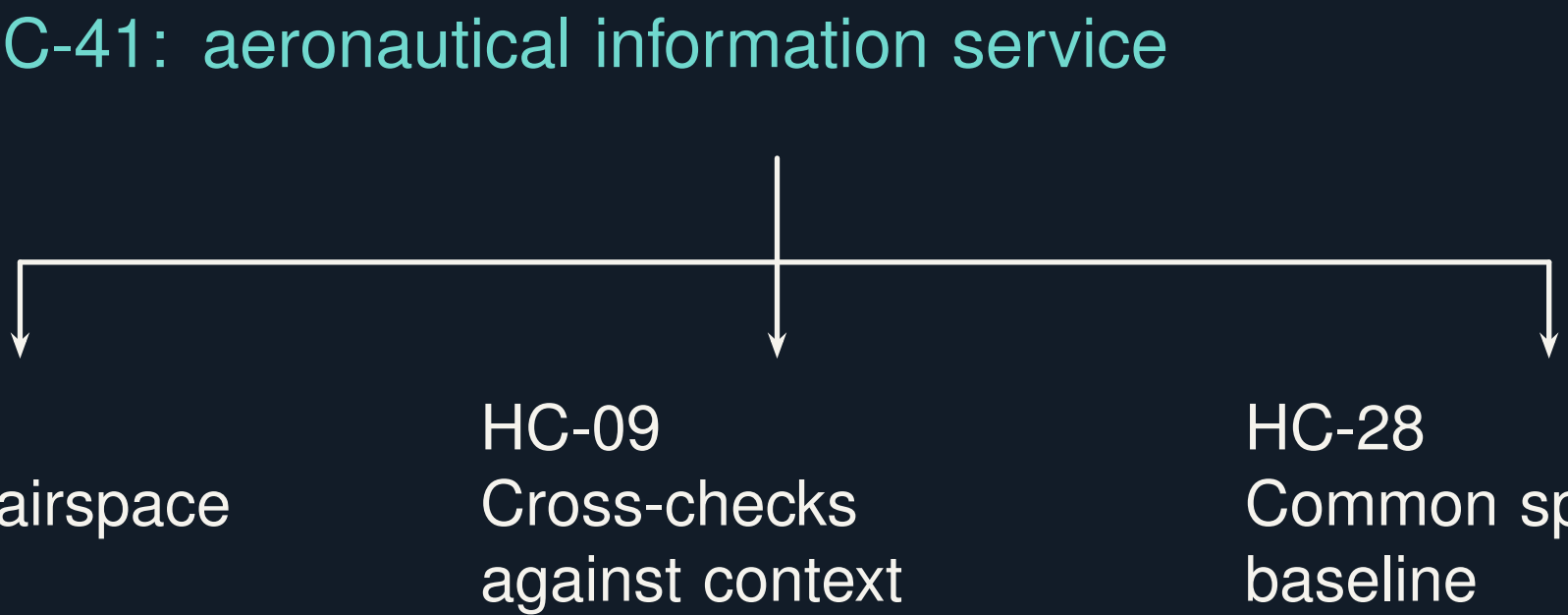
After launch: C2

Monitor separation and intervene in time

Allow for uncertainty and changes in trajectories.

Common cause analysis found shared services dependencies

One service loss can weaken several controls



HC-09

Cross-check against flight plan, mission, and sensor evidence

Derived requirement

Candidate air tracks shall be cross-checked against mission plans, published routes, scheduled windows and available sensor corroboration before hostile status is asserted.

Architectural elements

CID Manager
Airspace Coordination Manager
Decision Support & Display Service

V&V objective

Verify through scenario testing that route, procedure, and sensor context influence exclusion or inhibit logic.

People are the glue in a system of systems

People

Shared understanding
Coordination
Judgement / authority

Organisation

Aligned plans
Common procedures
Cross-unit coordination

Training

Joint exercises
Degraded conditions
Rehearsed coordination

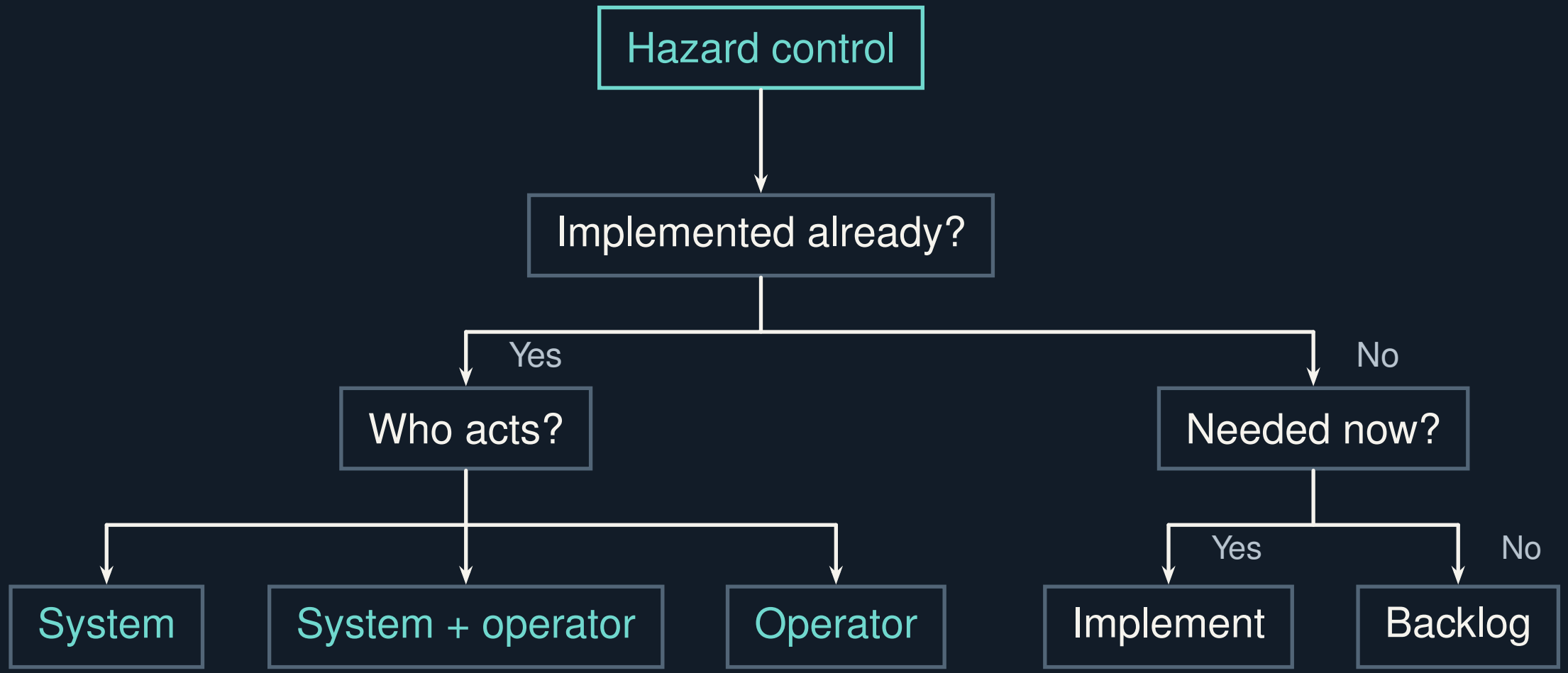
Background conditions supporting A–C2;
evaluate under realistic operating conditions.

4+1 case studies

Support development and validation of the model
by testing its fitness for purpose
not absolute correctness

Control effectiveness and risk reduction
still require evidence

And then integrate your results into an Agile program



- Audit behaviour
- Targeted V&V
- Log issues / risks

- Workflow / HMI
- Usability
- Log issues / risks

- Procedures / training
- Identify support
- Log issues / risks

What we can now say about managing fratricide risk

1. Existing hazard analysis tools can model fratricide at the SoS functional level

2. The model provides a traceable basis for deriving and allocating 44 controls

3. Case evidence and consistency checks can be used to verify the model and analysis

Control effectiveness still needs to be demonstrated.

Backup: case evidence

Vincennes, 1988	Shared airspace and track / IFF confusion
Black Hawk, 1994	Friendly information failed to reach shooters
Tornado, 2003	Recovery route, local operation, degraded ID
Gettysburg, 2024	Degraded services and coordination
F/A-18, 2003	Supplementary case: other-target exposure

Backup: 44 controls = 8 organisational / assurance + 1 service + 35 functional

Data and time integrity, with provenance

Common reference frames and data

Current airspace and friendly-position information

Inhibition when confidence is inadequate

Independent safe-hold and abort paths

Enterprise integration and joint training

Backup: barrier A fault-tree logic

A fails if A1 OR A2 OR A3 OR A4 occurs

A1: picture divergence	HC-05 failure AND HC-06 failure
A2: context not current	HC-07 failure OR HC-09 failure
A3: coordination failure	HC-26 failure
A4: spatial mismatch	HC-28 failure OR HC-29 failure

HC-07, HC-09 and HC-28 share exposure to HC-41 service loss.